

Quiz ²~~X~~ (Version 1)

CAS CS 132: *Geometric Algorithms*

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- ▷ You will have approximately 35 minutes to complete this exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

1 Matrix Equations

Determine a general form solution for the following matrix equation.

$$\begin{bmatrix} 1 & -3 & -10 \\ 0 & -1 & -4 \\ -1 & 2 & 6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 8 \\ 5 \\ -3 \end{bmatrix} - \begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 9 \\ 2 \\ -7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & -10 & 9 \\ 0 & -1 & -4 & 2 \\ -1 & 2 & 6 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -10 & 9 \\ 0 & 1 & 4 & -2 \\ 0 & -1 & -4 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.

$$x_1 = 3 - 2x_3$$

$$x_2 = -2 - 4x_3$$

x_3 is free

2 Linear Independence

Determine if the following vectors are linear independent. Show your work and justify your answer.

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -3 \\ -1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ -7 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ -3 & -5 & -7 \\ -1 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 3 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

Solution.

Lin. Ind. 1 pivot per column

3 Linear Equations and Span

Determine if $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ is the same as the solution set of the given linear equation. Show your work and justify your answer.

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 5 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \quad 3x_1 - 4x_2 + x_3 = 1$$

$$3(0) - 4(1) + 5 = 1$$

$$3(1) - 4(1) + 2 = 1$$

$$\text{so } \vec{v}_1, \vec{v}_2 \in \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : 3x_1 - 4x_2 + x_3 = 1 \right\} \quad \downarrow S$$

$$\text{but } \vec{0} \notin S \quad \text{and} \quad \vec{0} \in \text{span}\{\vec{v}_1, \vec{v}_2\}$$

Solution.

$$\text{No. } \vec{0} \in \text{span}\{\vec{v}_1, \vec{v}_2\} \quad \text{and}$$

$$\vec{0} \notin \text{sol. set of } 3x_1 - 4x_2 + x_3 = 1$$